

Transformer

Encoder only → context vector → best layer

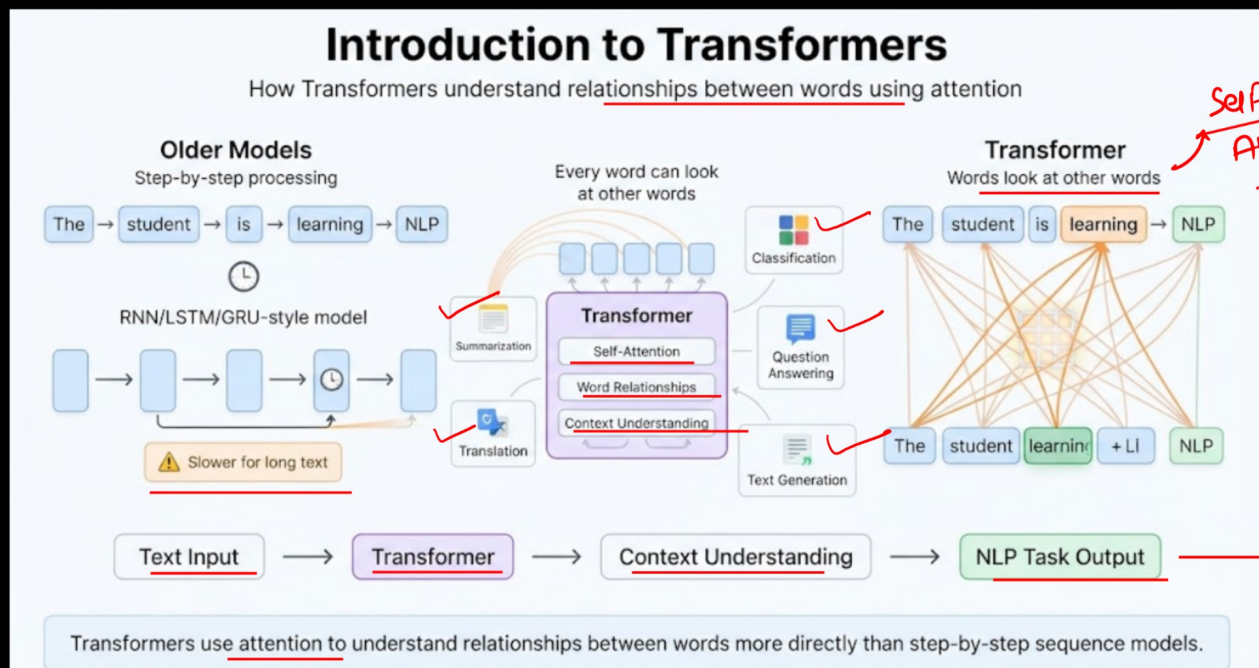
Hyperparameter BERT (12-encoders & 24-encoders)
↳ best-base

Decoder only

GPT (decoder only)

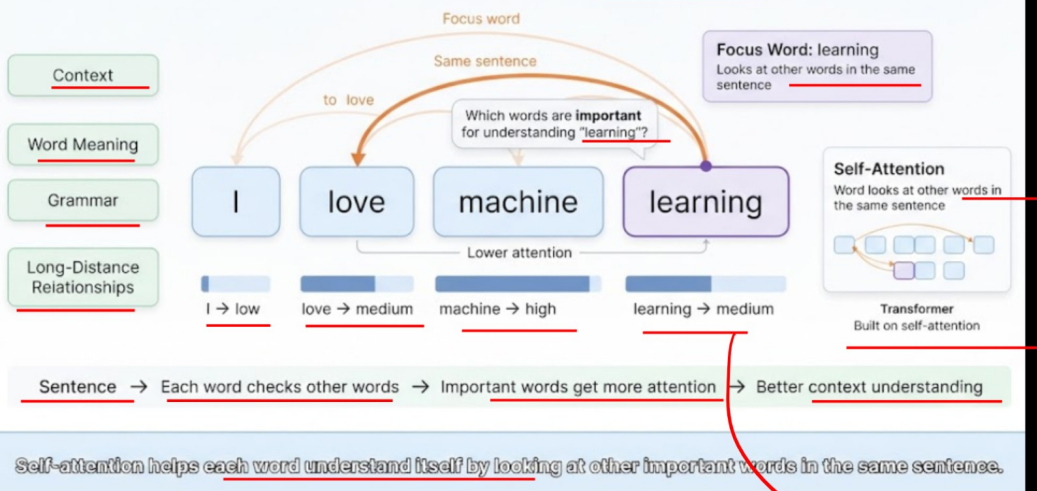
Deep Learning

Encoder → Self attention → feed forward
Neural
Network



Core Idea of Self-Attention

How each word looks at other words in the same sentence to understand meaning

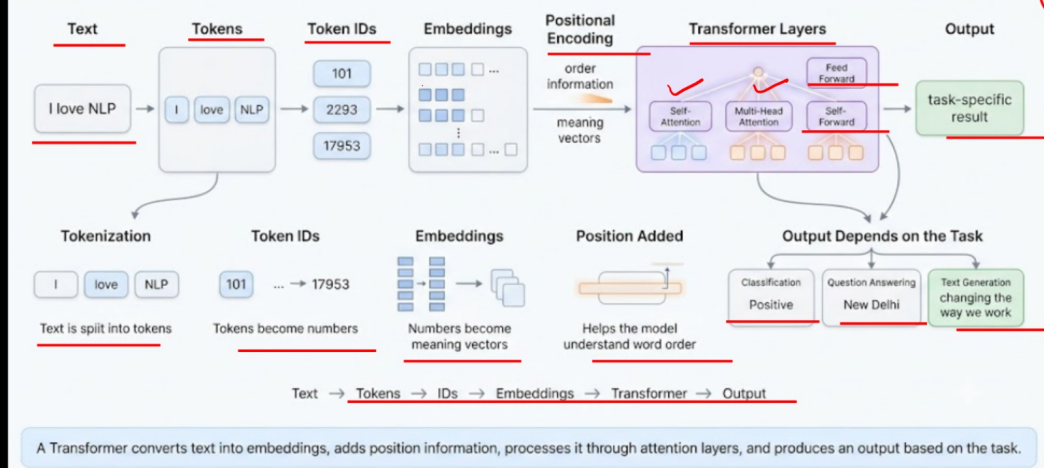


Attention score

$$\text{self attention}(z) = \text{softmax} \left(\frac{Q \cdot K^T}{\sqrt{d_k}} \right) \cdot V$$

Transformer Input and Output Flow

How text becomes tokens, embeddings, Transformer layers, and task-specific output



1 Grammar

2 Long range dependency

Transformers → Multi-head attention

Each head have its own

Head 1 → subject-verb

relationship

Q, K & V vectors

Head 2 → Pronouns

Head 3 $\rightarrow$ semantic meaning

Head 4 $\rightarrow$ Long Range Dependency

Decoder

Input $\rightarrow$ हम AI परीक्षा है!

Output $\rightarrow$ I Love

Language Modeling

masking
Predict the
next
token

GPT $\rightarrow$ Language Modeling

Self Supervised Learning

Pretraining

Publicly

✓ (actual)
predicted

output Embedding $\rightarrow$ result of the
Encoder Layer
(context vector)